Chapter 6: Mixed Strategies

ECON 5001
Prof. Paul J. Healy
Based on: *Game Theory* by Giacomo Bonanno

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Where We Left Off

Chapter 2: some games have **no** Nash equilibrium

- Matching Pennies, penalty kicks, play-calling...
- Being predictable gets you exploited

Chapter 5: **expected utility** ranks lotteries

- $u(o)$: outcome utility $U(L)$: expected utility

$$U(L) = \sum_i p_i \, u(o_i)$$

Our goal:

1. Let players **randomize**
2. Use expected utility to evaluate strategies in the face of a random opponent
3. Redefine equilibria to include random play
    - Show that, with this definition, it always exists

MobLab: “Matrix: Instructor Specified” ×2

- MobLab time!
- You’ll be paired with someone. Row player or Column player
- Two rounds, new partner each round. Each round played once

Round 1

Row	Column: Left	Column: Right
Top	80, 40	40, 80
Bottom	40, 80	80, 40

Round 2

Row	Column: Left	Column: Right
Top	**320**, 40	40, 80
Bottom	40, 80	80, 40

Feel free to talk! And play it the way you’d actually play it.

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Bottom	40, 80	80, 40

Who should respond to 320? Row? Column? Both?

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Round 2

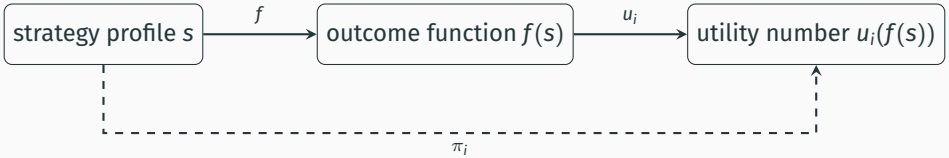
Row		Column: Left	Column: Right
	Top	**320**, 40	40, 80
	Bottom	40, 80	80, 40

Who should respond to 320? Row? Column? Both?
Paradoxical prediction: **Column** responds, **not** row

Strategic-Form Games With Randomness

Strategic-Form Games

Same as Chapter 2:



- Payoff $\pi_i(s)$ gives i's utility for the outcome generated by s
  - No need to specify outcomes, just payoffs
- Utility numbers are **expected utility values**, like little $u(o_i)$
  - Remember: only linear rescaling is allowed
- Bonanno: "games with cardinal payoffs"

How To Evaluate Randomness?

	Rock	Paper	Scissors
Rock	*Draw*	*Lose*	*Win*
Paper	*Win*	*Draw*	*Lose*
Scissors	*Lose*	*Win*	*Draw*

P1's outcome shown

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P1's outcome shown

- Player 1: Win $\succ$ Draw $\succ$ Lose

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- Let's try: $u(Win) = 1$, $u(Draw) = 0$, $u(Lose) = -1$

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P1's outcome shown

- Player 1: Win $\succ$ Draw $\succ$ Lose
- Let's try: $u(Win) = 1$, $u(Draw) = 0$, $u(Lose) = -1$
- And suppose your opponent is playing each strategy with $1/3$ chance

How To Evaluate Randomness?

	R (1/3)	P (1/3)	S (1/3)
Rock	0	−1	1
Paper	1	0	−1
Scissors	−1	1	0

P1's outcome utility ($u(o)$) shown

Use expected utility to evaluate strategies:

- $U(\text{Rock}) = (1/3) \cdot 0 + (1/3) \cdot (-1) + (1/3) \cdot 1 = 0$
- $U(\text{Paper}) = (1/3) \cdot 1 + (1/3) \cdot 0 + (1/3) \cdot (-1) = 0$
- $U(\text{Scissors}) = (1/3) \cdot (-1) + (1/3) \cdot 1 + (1/3) \cdot 0 = 0$

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You're exactly indifferent! All strategies are best replies.

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- $U(\text{Scissors}) = (1/3) \cdot (-1) + (1/3) \cdot 1 + (1/3) \cdot 0 = 0$

You're exactly indifferent! All strategies are best replies.
Thus, you're happy to randomize between them. ← **Indifference Principle**

Football Example: Run vs. Pass

Another source of randomness: **Random outcomes**

Example: Going for it on 4th down:

- Ohio State offense calls: **Run** or **Pass**
- Michigan's defense calls: **Stop Run** or **Stop Pass**
- Simultaneous: neither sees the other's call

Even after both calls, the play can work or fail

Outcome: a **lottery** over {Score, Fail}

Sometimes the Outcome Is a Lottery

	Stop Run	Stop Pass
Run	$\begin{pmatrix} \text{Score} & \text{Fail} \\ 0.4 & 0.6 \end{pmatrix}$	$\begin{pmatrix} \text{Score} & \text{Fail} \\ 0.8 & 0.2 \end{pmatrix}$
Pass	$\begin{pmatrix} \text{Score} & \text{Fail} \\ 0.9 & 0.1 \end{pmatrix}$	$\begin{pmatrix} \text{Score} & \text{Fail} \\ 0.3 & 0.7 \end{pmatrix}$

What is the payoff in each cell?

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What is the payoff in each cell? Use **expected utility**

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What is the payoff in each cell? Use **expected utility**

Normalize OSU's payoff: $u(\text{Score}) = 1$, $u(\text{Fail}) = 0$

$$U\begin{pmatrix} \text{Score} & \text{Fail} \\ 0.4 & 0.6 \end{pmatrix} = \underline{\qquad\qquad\qquad\qquad} = \Pr(\text{Score})$$

The Reduced Game

Multiply by 100 (linear rescaling). Payoff = % chance of each team's goal

		Michigan: Stop Run	Michigan: Stop Pass
Ohio State	Run	40, 60	80, 20
	Pass	90, 10	30, 70

Rule: If outcomes are random, use expected utility to calculate payoff

Convention: Skip the outcomes/lotteries, just write the payoff number in each cell

Bonanno §6.1 describes this in more detail. Skip it for this class.

Mixed Strategies

Any Nash Equilibrium?

Ohio State \ Michigan	Stop Run	Stop Pass
Run	40, 60	80, 20
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No cell has both underlined. No (pure) Nash equilibrium!

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What's optimal here?

Be unpredictable: **Randomize**

Any Nash Equilibrium?

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Pass	90, 10	30, 70

No cell has both underlined. No (pure) Nash equilibrium!

What's optimal here?

Be unpredictable: **Randomize**
But with what probabilities?? 50-50??

Best Reply to a Mixed Strategy

Suppose Michigan plays $\begin{pmatrix} \text{Stop Run} & \text{Stop Pass} \\ 0.8 & 0.2 \end{pmatrix}$. What should OSU do?

Ohio State \ Michigan	Stop Run	Stop Pass
Run	40, 60	80, 20
Pass	90, 10	30, 70

$\Pi_{OSU}(\text{Run},\ q = 0.8) =$

$\Pi_{OSU}(\text{Pass},\ q = 0.8) =$

If $q = 0.8$: Run payoff: _____, Pass payoff _____

OSU's best reply to $q = 0.8$: _____

Best Reply to a Mixed Strategy

Suppose Michigan plays $\begin{pmatrix} \text{Stop Run} & \text{Stop Pass} \\ q & 1-q \end{pmatrix}$. What should OSU do?

Ohio State \ Michigan	Stop Run	Stop Pass
Run	40, 60	80, 20
Pass	90, 10	30, 70

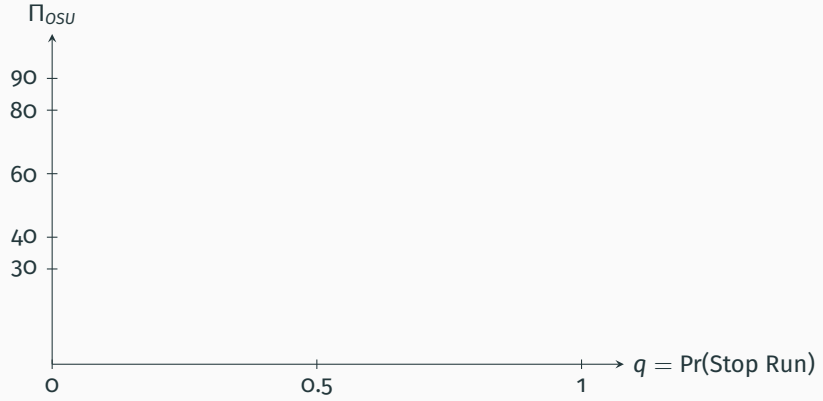
$\Pi_{OSU}(\text{Run},\ q) =$

$\Pi_{OSU}(\text{Pass},\ q) =$

If $q = 0.5$: Run gives _____, Pass gives _____

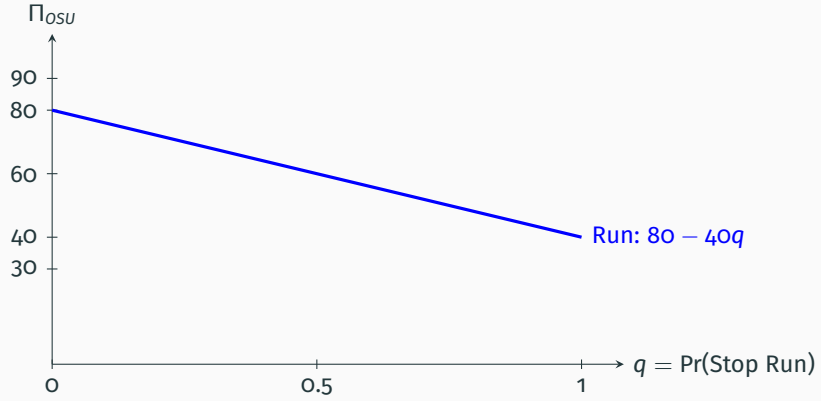
OSU's best reply: _____

Seeing It in a Graph



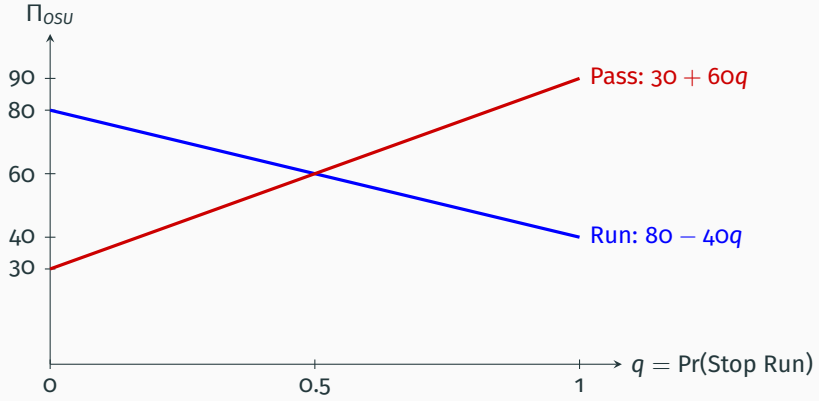
Plot each pure strategy's expected payoff against q

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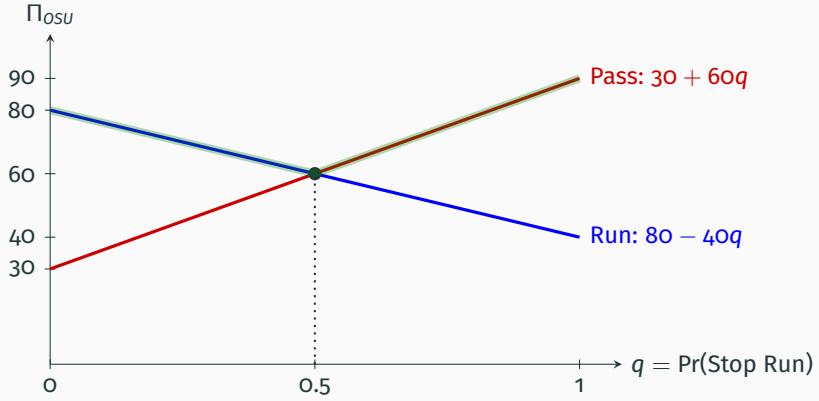
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Seeing It in a Graph



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Seeing It in a Graph



Best reply = the higher line. They cross at $q =$ _____

OSU's Best Reply

$$BR_{OSU}(q) = \begin{cases} \text{Run} & \text{if } q < \frac{1}{2} \\ \text{_____} & \text{if } q = \frac{1}{2} \\ \text{Pass} & \text{if } q > \frac{1}{2} \end{cases}$$

Fact: A mix of Run and Pass is a best reply **only** when Run and Pass tie

- Otherwise, shift weight to the better one!

Fact: When they tie, **every** mix is a best reply

In general: σ_i is a best reply $\iff$ every pure strategy it uses is a best reply

Definition: Mixed Strategy

A **mixed strategy** σ_i is a probability over each of player i's strategies

Ex: $\sigma_1 = \begin{pmatrix} \text{Run} & \text{Pass} \\ p & 1-p \end{pmatrix}$ $\quad$ $\sigma_2 = \begin{pmatrix} \text{Stop Run} & \text{Stop Pass} \\ q & 1-q \end{pmatrix}$

Notation:

- $\sigma_i(s_i)$: probability that σ_i puts on s_i. Rule: $\sum_{s_i} \sigma_i(s_i) = 1$
  - $\sigma_1(\text{Run}) = p$, $\sigma_1(\text{Pass}) = 1 - p$
  - $\sigma_2(\text{Stop Run}) = q$, $\sigma_2(\text{Stop Pass}) = 1 - q$
- Σ_i: set of all mixed strategies of player i. So $\sigma_i \in \Sigma_i$
- A non-mixed strategy $s_i \in S_i$ is called a a **pure strategy**
  - Pure strategy $=$ mixed strategy with probability 1 on one s_i

Assume: players randomize **independently** (no correlation)

Expected Payoff of a Mixed Profile

A mixed strategy profile $\sigma = (\sigma_1, \sigma_2)$ generates a lottery over the **cells**

Probability of cell $s = (s_1, s_2)$: $\sigma(s) = \sigma_1(s_1) \times \sigma_2(s_2)$ (independence)

Expected payoff: weight each cell's payoff by the chance of landing in it

$$\underbrace{\Pi_i(\sigma)}_{\text{expected payoff}} = \sum_{s \in S} \underbrace{\sigma(s)}_{\text{prob. of cell } s} \cdot \underbrace{\pi_i(s)}_{\text{payoff in cell } s}$$

Capital Π: payoff from a mixed profile. Small π: from a pure profile

Same idea as *U* vs. *u* in Chapter 5

Definition: Mixed-Strategy Nash Equilibrium

In words: every player's mix is a best reply to the other players' mixes

A mixed profile $\sigma^* = (\sigma_1^*, \ldots, \sigma_n^*)$ is a **Nash equilibrium** if, for every player i,

$$\underbrace{\Pi_i(\sigma_i^*, \sigma_{-i}^*)}_{\text{payoff from equilibrium mix}} \geq \underbrace{\Pi_i(\sigma_i, \sigma_{-i}^*)}_{\text{payoff from any other mix}} \qquad \text{for all } \sigma_i \in \Sigma_i$$

Equivalently: $\sigma_i^* \in BR_i(\sigma_{-i}^*)$ for every i

A pure-strategy NE is still a NE when mixing is allowed (Bonanno p. 211).

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Same as Chapter 2: $s_i^* \in BR_i(s_{-i}^*)$ for every i

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Nash's Theorem

Theorem (Nash's 1951 PhD Dissertation at Princeton)
Every finite game has **at least one** Nash equilibrium in mixed strategies.

Finite game: finitely many players, each with finitely many strategies.

- Matching Pennies: no pure NE, but a mixed one
- Run vs. Pass: no pure NE, but a mixed one

Proof uses "Kakutani's fixed-point theorem". (Not covered here.)

Recall: We assume players maximize expected utility.

Finding Mixed Equilibria

The Indifference Principle

Indifference Principle: If you mix between two strategies in equilibrium, they must have the same expected payoff. (If not, you'd switch to playing the better one for sure!)

If σ_i^* puts positive probability on both s_i and s_i', then

$$\underbrace{\Pi_i(s_i, \sigma_{-i}^*)}_{\text{payoff from } s_i} = \underbrace{\Pi_i(s_i', \sigma_{-i}^*)}_{\text{payoff from } s_i'} = \underbrace{\Pi_i(\sigma^*)}_{\text{equilibrium payoff}}$$

Implication:

1. If I'm mixing between two strategies in equilibrium, it's because I'm indifferent
2. Thus, you must be mixing in a way that **makes me indifferent**
3. And if you're mixing, I'm also **making you indifferent**

So, mixed strategy equilibrium $\Rightarrow$ designed to make both players **indifferent**

Bonanno Theorem 6.3.1

An Example

		P2 L	P2 C	P2 R
P1	T	1, 0	0, 1	2, −1
P1	M	0, 2	1, 0	2, −1
P1	B	−1, 0	−1, 0	1, 3

B is strictly dominated by *M*

An Example

		P2 L	P2 C	P2 R
P1	T	1, 0	0, 1	2, −1
P1	M	0, 2	1, 0	2, −1
P1	B	−1, 0	−1, 0	1, 3

Then R is strictly dominated by C.
So, reduce the game to $\{T, M\} \times \{L, C\}$

An Example

		P2 L	P2 C	P2 R
P1	T	1, 0	0, 1	2, −1
P1	M	0, 2	1, 0	2, −1
P1	B	−1, 0	−1, 0	1, 3

Suppose $Pr(L) = q$ and $Pr(C) = 1 - q$. (We know $Pr(R) = 0$)

When is P1 indifferent between T and M?

$$\Pi_1(T) = q(1) + (1-q)(0) = q$$

$$\Pi_1(M) = q(0) + (1-q)1 = 1 - q$$

$$\Pi_1(T) = \Pi_1(M) \Rightarrow q = 1 - q \Rightarrow q = 1/2$$

Prediction: P1 plays $q = 1/2$

An Example

		P2 L	P2 C	P2 R
P1	T	1, 0	0, 1	2, −1
P1	M	0, 2	1, 0	2, −1
P1	B	−1, 0	−1, 0	1, 3

Suppose $Pr(T) = p$ and $Pr(M) = 1 - p$. (We know $Pr(B) = 0$)

When is P2 indifferent between L and C?

$$\Pi_2(L) = p(0) + (1-p)(2) = 2 - 2p$$

$$\Pi_2(C) = p(1) + (1-p)0 = p$$

$$\Pi_2(T) = \Pi_2(M) \Rightarrow p = 2 - 2p \Rightarrow 3p = 2 \Rightarrow p = 2/3$$

Prediction: P2 plays $p = 2/3$

Recipe: Mixed NE in a 2×2

1. Run IDSDS first. Dominated strategies get probability 0
2. Underline. Note any pure NE
3. P2 plays q, $1 - q$. Find the q that makes **P1** indifferent
   - **Notice:** P2 is making P1 indifferent!
4. P1 plays p, $1 - p$. Find the p that makes **P2** indifferent
   - **Notice:** P1 is making P2 indifferent!
5. Both strictly between 0 and 1? That's a mixed NE

Solving the Two-Point Game

Ohio State	Michigan: Stop Run	Michigan: Stop Pass
Run	40, 60	80, 20
Pass	90, 10	30, 70

Step 1: Michigan's q makes OSU indifferent

$q^* =$

Solving the Two-Point Game

Ohio State \ Michigan	Stop Run	Stop Pass
Run	40, 60	80, 20
Pass	90, 10	30, 70

Step 2: OSU's p makes Michigan indifferent

$p^* =$

Mixed NE:

OSU scores _____% of the time

The Unsettling Part of MSNE

Consider Rock Paper Scissors (RPS) whose MSNE is $\sigma^* = ((1/3, 1/3, 1/3), (1/3, 1/3, 1/3))$

- In the MSNE you're indifferent between Rock, Paper, and Scissors.
- But then why do you play $(1/3, 1/3, 1/3)$?? You have **no incentive** either way!

Nash reasoning:

- When considering deviating, you assume **they don't change** their strategy σ^*_{-i}.
- If they're fixed at $(1/3, 1/3, 1/3)$ then you don't care what you play.
- Deviating away from $(1/3, 1/3, 1/3)$ doesn't help or hurt you.

"Dynamic" reasoning:

- If you deviate away from $(1/3, 1/3, 1/3)$ then they'll respond by moving away from $(1/3, 1/3, 1/3)$ to take advantage of you
- ...and then you'd change your strategy, and then so would they...
- Only "stable" point: Mixed Strategy Nash equilibrium σ^*
  - Caveat: that dynamic process won't *converge* to σ^*... it'll cycle forever
  - Stable equilibrium vs. unstable equilibrium.

Back to MobLab

Round 1

Row \ Column	Left	Right
Top	80, 40	40, 80
Bottom	40, 80	80, 40

Round 2

Row \ Column	Left	Right
Top	320, 40	40, 80
Bottom	40, 80	80, 40

Round 1 mixed NE:

Round 2 mixed NE:

Only Row's payoff changed. Whose mix changes in the NE??

Back to MobLab

	% Rows playing Top		% Columns playing Left	
	Round 1	Round 2	Round 1	Round 2
Nash prediction	50	50	50	12.5
Goeree & Holt (2001)	$\approx$ 50	96	$\approx$ 50	16

Lab result: Rows chase the 320. NE says Row's own payoffs shouldn't matter! Columns anticipate it and play Left even less

Not in the book. Source: Goeree & Holt (2001), "Ten Little Treasures of Game Theory and Ten Intuitive Contradictions," *American Economic Review* 91(5).

Pure and Mixed Together

The Sunday Morning Game Revisited

Alex and Blair, dead phones: Buckeye Donuts or HangOverEasy?

		Blair: BuckyDs	Blair: HangOE
Alex	BuckyDs	4, 3	2, 1
Alex	HangOE	1, 2	3, 4

Now read the payoffs as **cardinal** (outcome utilities)

The Sunday Morning Game Revisited

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Pure NE: _____

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Now read the payoffs as **cardinal** (outcome utilities)

Pure NE: _____

Is there also a mixed NE?

Sunday Morning: The Mixed NE

Alex \ Blair	BuckyDs	HangOE
BuckyDs	4, 3	2, 1
HangOE	1, 2	3, 4

Blair's q makes Alex indifferent:

Alex's p makes Blair indifferent:

Payoffs at the mixed NE: Alex _____ Blair _____ Better or worse than the pure NE?

Indifference Does Not Imply Nash Equilibrium

With more than 2 strategies, indifference $\not\Rightarrow$ Nash equilibrium

Player 1 \ Player 2	D (½)	E (½)	
A (½)	3, 0	0, 2	$\frac{1}{2}(3)+\frac{1}{2}(0)=1.5$
B (½)	0, 2	3, 0	$\frac{1}{2}(0)+\frac{1}{2}(3)=1.5$
C (0)	2, 0	2, 1	$\frac{1}{2}(2)+\frac{1}{2}(2)=2$

1.5 BR

B, R

Try: P1 plays $\begin{pmatrix} A & B & C \\ \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}$, P2 plays $\begin{pmatrix} D & E \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$

P1 indifferent between A and B? $\Pi_1(A) =$ 1.5 $\Pi_1(B) =$ 1.5

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With more than 2 strategies, indifference $\not\Rightarrow$ Nash equilibrium

Player 1 \ Player 2	D	E
A	3, 0	0, 2
B	0, 2	3, 0
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Try: P1 plays $\begin{pmatrix} A & B & C \\ \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}$, P2 plays $\begin{pmatrix} D & E \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$

P1 indifferent between A and B? $\Pi_1(A) =$ 1.5 $\Pi_1(B) =$ 1.5

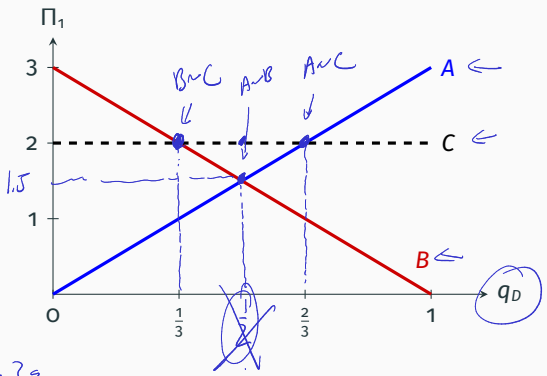
But $\Pi_1(C) =$ 2 So mixing 50-50 over A and B is **not** a NE!

Lesson: Need indifference **and** no better strategy

Best Replies With Three Strategies

Player 1 \ Player 2	D	E
A	3, 0	0, 2
B	0, 2	3, 0
C	2, 0	2, 1

Π_1



$\Pr(D) = q_D$

$\Pr(E) = q_E = 1 - q_D$

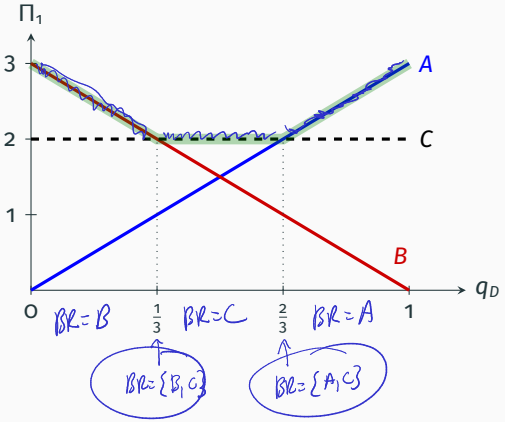
$\Pi_1(A, q_D) = 3q_D$

$\Pi_1(B, q_D) = 3(1 - q_D) = 3 - 3q_D$

$\Pi_1(C, q_D) = 2$

Best Replies With Three Strategies

Player 1 \ Player 2	D	E
A	3, 0	0, 2
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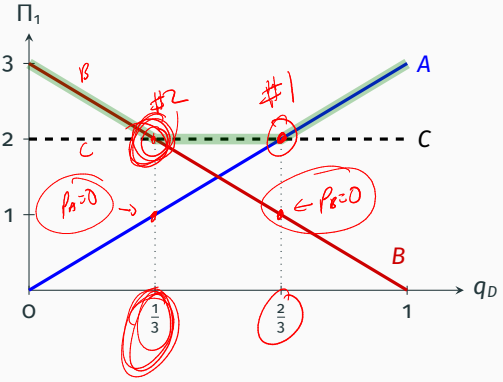
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$\Pi_1(A, q_D) = 3q_D$
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$\Pi_1(C, q_D) = 2$

$BR_1(q_D)$: B if $q_D < \frac{1}{3}$ C if $\frac{1}{3} < q_D < \frac{2}{3}$ A if $q_D > \frac{2}{3}$

Best Replies With Three Strategies

Player 1 \ Player 2	D	E
A	3, 0	0, 2
B	0, 2	3, 0
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$\Pr(D) = q_D$
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$\Pi_1(A, q_D) = 3q_D$
$\Pi_1(B, q_D) = 3(1 - q_D) = 3 - 3q_D$
$\Pi_1(C, q_D) = 2$

No pure strategy NE $\Rightarrow$ Must be mixing $\Rightarrow$ q_D needs to make P1 indifferent
Possible solution #1: Set $q_D = 2/3$, so P1 mixes between A & C, setting $p_B = 0$.
Possible solution #2: Set $q_D = 1/3$, so P1 mixes between B & C, setting $p_A = 0$.

What is P1's MSNE Strategy?

How should P1 mix in the MSNE??

Player 1 \ Player 2	D	E
A	3, 0	0, 2
B	0, 2	3, 0
C	2, 0	2, 1

$\Pr(A) = p_A$

$\Pr(B) = p_B$

$\Pr(C) = p_C = 1 - p_A - p_B$

P1 MUST make P2 indifferent:

$\Pi_2(\underline{D}, (p_A, p_B, p_C)) = 2p_B$

$\Pi_2(\underline{E}, (p_A, p_B, p_C)) = 2p_A + 1p_C =$

$2p_A + (1 - p_A - p_B) = 1 + p_A - p_B$

Make P2 indifferent (cont.):

$2p_B = 1 + p_A - p_B$

$\mathbf{3p_B = 1 + p_A}$ ← P1 should satisfy this

What is P1's MSNE Strategy?

How should P1 mix in the MSNE??

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Make P2 indifferent (cont.):
$2p_B = 1 + p_A - p_B$
$\mathbf{3p_B = 1 + p_A}$ ← P1 should satisfy this

Recall the 2 possible solutions...
Try possible solution #1: $p_B = 0$:
$3 \cdot 0 = 1 + p_A$
$p_A = -1$ IMPOSSIBLE!

What is P1's MSNE Strategy?

How should P1 mix in the MSNE??

Player 1 \ Player 2	D	E
A	3, 0	0, 2
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Make P2 indifferent (cont.):
$2p_B = 1 + p_A - p_B$
$\mathbf{3p_B = 1 + p_A}$ ← P1 should satisfy this

Recall the 2 possible solutions...
Try possible solution #1: $p_B = 0$:
$3 \cdot 0 = 1 + p_A$
$p_A = -1$ IMPOSSIBLE!

Try possible solution #2: $p_A = 0$:
$3 \cdot p_B = 1 + 0$
$p_B = 1/3$ POSSIBLE!

What is P1's MSNE Strategy?

How should P1 mix in the MSNE??

		Player 2 D	Player 2 E
Player 1	A	3, 0	0, 2
Player 1	B	0, 2	3, 0
Player 1	C	2, 0	2, 1

$\Pr(A) = p_A$
$\Pr(B) = p_B$
$\Pr(C) = p_C = 1 - p_A - p_B$

P1 MUST make P2 indifferent:
$\Pi_2(D, (p_A, p_B, p_C)) = 2p_B$
$\Pi_2(E, (p_A, p_B, p_C)) = 2p_A + 1p_C =$
$2p_A + (1 - p_A - p_B) = 1 + p_A - p_B$

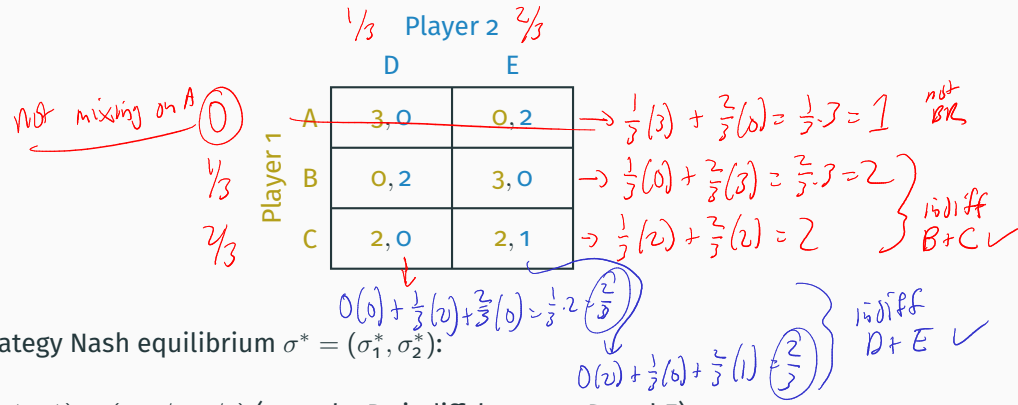
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Recall the 2 possible solutions...
Try possible solution #1: $p_B = 0$:
$3 \cdot 0 = 1 + p_A$
$p_A = -1$ IMPOSSIBLE!

Try possible solution #2: $p_A = 0$:
$3 \cdot p_B = 1 + 0$
$p_B = 1/3$ POSSIBLE!

Solution for P1:
$p_A = 0$
$p_B = 1/3$
$p_C = 2/3$ since $p_A + p_B + p_C = 1$

Best Replies With Three Strategies



Mixed-Strategy Nash equilibrium $\sigma^* = (\sigma_1^*, \sigma_2^*)$:

$\sigma_1^* = (p_A^*, p_B^*, p_C^*) = (0, 1/3, 2/3)$ (to make P2 indiff. between D and E)

$\sigma_2^* = (q_D^*, q_E^*) = (1/3, 2/3)$ (to make P1 indiff. between B and C)

Let's write the expected payoffs next to each strategy to verify...

What Does Mixing Mean?

At a mixed NE you're **indifferent**. So why mix exactly 1/3???

It's **not** because of any strict incentive to do so...

it's just the only "stable state" of the system.

Equivalently: at any *other* strategy profile, someone **will** deviate.

Bonanno calls mixed NE "conceptually problematic." Most agree.

Mixing in the Real World

Lots of situations boil down to “be unpredictable”:

- **Penalty kicks.** Kick left or right? Dive left or right?
- **Tennis serves.** To the forehand or the backhand?
- **Play-calling, pitch selection.** Run/pass. Fastball/curveball
- **Poker.** Bluff with a bad hand, or fold?
- **Inspections.** Tax audits, parking tickets, drug tests, fare checks
- **Security.** LAX randomizes its checkpoints and patrols

Lab/field result: Pro soccer and tennis players mix close to NE rates

Not in the book. Sources: Walker & Wooders (2001, *AER*); Chiappori, Levitt & Groseclose (2002, *AER*); Palacios-Huerta (2003, *REStud*); Pita et al. (2008, AAMAS).

Complications In Real Examples

Real settings have:

- More than two strategies
- Repeated play: patterns can be spotted and exploited
- Players differ in skill (a left-footed kicker)
- Players don’t payoffs of others, can’t make indifferent

Core idea survives: if you’re predictable, you get exploited

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- Players don’t payoffs of others, can’t make indifferent
  - Part IV of Bonanno: Games of “Incomplete information”

Core idea survives: if you’re predictable, you get exploited

Dominance With Mixed Strategies

Remember This One?

From Chapter 2: M is never a best reply to a **pure** strategy, but isn't dominated

Player 1 \ Player 2	L	C	R
U	4, 1	2, 0	4, 2
M	3, 2	3, 1	3, 0
D	0, 0	4, 2	0, 1

Suppose P2 plays $\begin{pmatrix} L & C & R \\ 0.4 & 0.6 & 0 \end{pmatrix} = \sigma_2$

$\Pi_1(U) = 2.8$ $\quad \Pi_1(M) = 3$ $\quad \Pi_1(D) = 2.4$

M is a BR to σ_2

Theorem: never BR to any σ_2 ⟺ dominated
BR to some σ_2 ⟺ not dominated

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Player 1 \ Player 2	L	C	R
U	4, 1	2, 0	4, 2
M	3, 2	3, 1	3, 0
D	0, 0	4, 2	0, 1

Suppose P2 plays $\begin{pmatrix} L & C & R \\ 0.4 & 0.6 & 0 \end{pmatrix}$

$\Pi_1(U) =$ $\Pi_1(M) =$ $\Pi_1(D) =$

M **is** a best reply, to a **mixed** strategy!

Dominance vs. Never Best Reply

We know that “strictly dominated $\Rightarrow$ never a best reply to pure strategies”

What about the other direction? Does “never best reply to P.S.” $\Rightarrow$ strictly dominated? No

- No! It can still be a B.R. to some mixed strategy

Dominated by a Mix

Player 1 \ Player 2	D	E
A	0, 1	4, 0
B	1, 2	1, 4
C	2, 0	0, 1

Is B dominated by A? By C? No.

Try the mix $\begin{pmatrix} A & B & C \\ \frac{1}{3} & 0 & \frac{2}{3} \end{pmatrix}$: vs. D: $\frac{4}{3}$ vs. E: $\frac{4}{3}$

σ_1 strictly dominates playing B

IDSDS: remove B

Dominated by a Mix

Player 1 \ Player 2	D	E
A	0, 1	4, 0
B	1, 2	1, 4
C	2, 0	0, 1

Is B dominated by A? By C?

Try the mix $\begin{pmatrix} A & B & C \\ \frac{1}{3} & 0 & \frac{2}{3} \end{pmatrix}$: vs. D: vs. E:

B is **strictly dominated by a mixed strategy**
So B is never a best reply, to anything P2 might do

Dominated By a Mix

Lesson:

- It's hard to see what's actually dominated
- Could be dominated by a pure strategy...
- Could be dominated by a mixed strategy ← harder to identify

But there's a useful link between "never a best reply to any mixed strategy of the opponent" and "strictly dominated"

Pearce's Theorem

Theorem (Pearce, 1984)
In a two-player game, s_i is never a best reply to any mixed strategy of the opponent

$\iff$ if and only if

s_i is strictly dominated by some mixed strategy of player i

- Fixes Chapter 2's gap: "never a best reply" = "dominated by a mix"

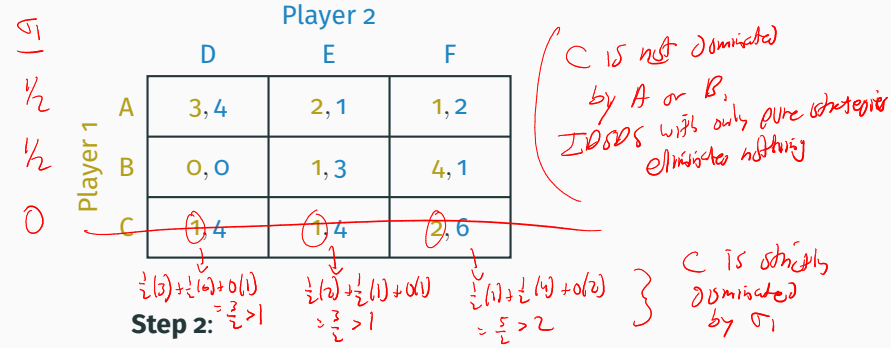
"Never BR (including mixed actions)" is same as "strictly dominated"

Three or more players: true if beliefs may be correlated (Bonanno §6.4).

Updated Version of IDSDS: Cardinal IDSDS aka "Rationalizability"

Same as IDSDS, but also delete strategies dominated by **mixes**

- Survivors are called **rationalizable** strategies



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$\sigma_2 = (\frac{1}{2}, \frac{1}{2}, 0)$

Player 1 \ Player 2	D	E	F
A	3, 4	2, 1	1, 2
B	0, 0	1, 3	4, 1
C	1, 4	1, 4	2, 6

σ_2 F

$\frac{1}{2}(4) + \frac{1}{2}(1) + 0(2) = \frac{5}{2} > 2$

$\frac{1}{2}(0) + \frac{1}{2}(3) + 0(1) = \frac{3}{2} > 1$

F dom. by σ_2

Step 1: C dom. by $(\frac{1}{2}, \frac{1}{2}, 0)$ **Step 2**:

Step 3: **Step 4**:

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Same as IDSDS, but also delete strategies dominated by **mixes**

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Step 1: C dom. by $(\frac{1}{2}, \frac{1}{2}, 0)$ **Step 2**: F dom. by $(\frac{1}{2}, \frac{1}{2}, 0)$

Step 3: B dom by A **Step 4**:

Updated Version of IDSDS: Cardinal IDSDS aka “Rationalizability”

Same as IDSDS, but also delete strategies dominated by **mixes**

- Survivors are called **rationalizable** strategies



Step 1: *C* dom. by $(\frac{1}{2}, \frac{1}{2}, 0)$

Step 2: *F* dom. by $(\frac{1}{2}, \frac{1}{2}, 0)$

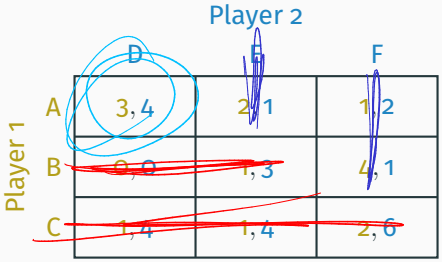
Step 3: *B* dom. by *A*

Step 4: E dom by D

Updated Version of IDSDS: Cardinal IDSDS aka “Rationalizability”

Same as IDSDS, but also delete strategies dominated by **mixes**

- Survivors are called **rationalizable** strategies



Step 1: C dom. by $(\frac{1}{2}, \frac{1}{2}, 0)$

Step 2: F dom. by $(\frac{1}{2}, \frac{1}{2}, 0)$

Step 3: B dom. by A

Step 4: E dom. by D

Updated Version of IDSDS: Cardinal IDSDS

New solution concept: Rationalizable strategies

Player 1 \ Player 2	D	E	F
A	3, 4	2, 1	1, 2
B	0, 0	1, 3	4, 1
C	1, 4	1, 4	2, 6

against pure strategies only

IDSDS: $\{A, B, C\} \times \{D, E, F\}$ (nothing is eliminated)

Rationalizable strategies: $\{(A, D)\}$ (unique prediction)

IDSDS, allowing dominance by mixed strategies as well

New solution concept: RATIONALIZABILITY (IDSDS including mixed strategies of opponent)

Rationalizability $\subseteq$ IDSDS

Let’s Work One Out

Gameday Parking

Jordan drives to campus for the Michigan game. Three options:

- **Pay** for a gameday lot
- Park illegally in a **West** Campus lot
- Park illegally on **Neil** Ave

CampusParc has one tow crew. It can patrol West Campus or Neil Ave

Jordan \ CampusParc	Patrol West	Patrol Neil
Pay	1, 0	1, 0
West	0, 2	4, 0
Neil	3, 0	0, 1

Gameday Parking

$q_W = 4/7$ $q_N = 1 - q_W = 3/7$

CampusParc

Jordan	Patrol West	Patrol Neil	
Pay	1, 0	1, 0	→ 1
West	0, 2	4, 0	→ $0q_W + 4(1-q_W) = 4 - 4q_W$
Neil	3, 0	0, 1	→ $3q_W + 0(1-q_W) = 3q_W$

$0 = p_P$

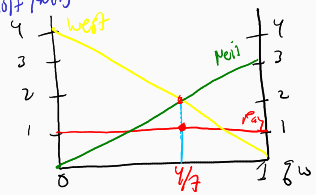
$1/3 = p_W$

$1 - p_P - p_W \quad 2/3 = p_N$

mix

$P_P(0) + P_W(2) + P_N(0) = 2p_W$

$P_P(0) + P_W(0) + P_N(1) = p_N = 1 - p_P - p_W$



patrol west patrol Neil

P2 is diff: $2p_W = 1 - p_P - p_W$

$3p_W = 1 - p_P$

But we know $p_P = 0$

$3p_W = 1 - 0$

$3p_W = 1$

$p_P = 0$ $p_W = 1/3$ $p_N = 2/3$

P1 payoff at $q_W = 4/7$:

Pay: 1

West: $4 - 4\left(\frac{4}{7}\right) = 4 - \frac{16}{7} = \frac{28-16}{7} = \frac{12}{7} > 1$

Neil: $3\left(\frac{4}{7}\right) = \frac{12}{7} > 1$

West ~ Neil

$4 - 4q_W = 3q_W$

$4 = 7q_W$

$q_W = 4/7$ $q_N = 1 - 4/7 = 3/7$

Gameday Parking

1. Is any strategy strictly dominated by a **pure** strategy? NO
2. Is any strategy strictly dominated by a **mixed** strategy? Delete it Yes, Pay is dom.
3. Is there a pure-strategy NE? No
4. Find the mixed NE and both players' expected payoffs.
   Check that Jordan really doesn't want to Pay

   $\sigma_1^* = (0, 1/3, 2/3)$ $\sigma_2^* = (\frac{4}{7}, \frac{3}{7})$

   Yes, Jordan would pay vs. $\sigma_2 = 4/7$
5. CampusParc now earns 4 (not 2) for a tow at West Campus.
   Whose mix changes? Does Jordan's payoff change?

   P1 changes, not P2

What Have We Learned, Charlie Brown?

- Lottery in a cell? Its expected utility is the payoff
- **Mixed strategy**: a lottery over your own pure strategies
- $\Pi_i(\sigma)$: weight each cell's payoff by its probability
  - Needs **cardinal** payoffs. That's why we did Chapter 5
- **Mixed NE**: every mix is a best reply to the others
- **Nash (1951)**: every finite game has one
- **Indifference principle**: my mix makes *you* indifferent
  - Necessary, not sufficient. Check unused strategies
- **Pearce**: never a best reply $\iff$ dominated by a mix

Next: mixing in game trees (behavioral strategies)

Reading & Practice

- Bonanno, *Game Theory*, Chapter 6
  - §6.1: skim. Lotteries in cells, done formally
  - §6.2–6.4: read carefully
- Exercises: §6.5.2 (6.3, 6.4), §6.5.3 (6.7, 6.8, 6.10), §6.5.4 (6.12)
  - 6.13 and 6.14 are harder. 6.15 (the dog in the river) is a challenge
- Solutions in §6.6. Try them first